

standalone mode, hence obviating the necessity for the former. Hadoop provides only two functions to Spark—processing by MapReduce and storage using the Hadoop Distributed File System (HDFS). Spark doesn't replace Hadoop as the two aren't mutually exclusive. Instead, they complement each other and result in an extremely powerful model.

The power of the Apache Spark engine

Speed: Spark uses in-memory cluster processing, which means it reduces the I/O operations for iterative algorithms as it stores the intermediate data generated in the memory instead of writing it back to the disk. Data can be stored on the RAM of the server machine and, hence, runs 100x quicker in memory and up to 10x faster on disk as compared to Hadoop. Moreover, due to its bottom-up engineering and the usage of RDDs, the fundamental data structure of Spark allows transparent storage of data in memory and persistence to disk only when it's needed. 'Lazy evaluation' is a feature that also contributes to Spark's speed by delaying the evaluation of any expression or operation until the value is needed by another expression. This avoids repeated evaluation of the same expression, and allows the definition of control flow and potentially infinite sets.

Libraries: Spark is equipped with standard built-in high-level libraries, including compatibility with SQL queries (SparkSQL), machine learning (MLlib), and streaming data and graph processing (GraphX), in addition to the simple MapReduce functionalities of the MR model. These increase the productivity of developers by allowing them to use the functionalities in fewer lines of code, yet create complex workflows. Spark is compatible with real-time processing applications.

Multiple languages: Programmers have the advantage of coding in familiar languages as Spark provides stable APIs in Java, Scala, Python, R and SQL. The Spark SQL component allows the import of structured data and its integration with unstructured data from other sources. Spark has over 100 high-level operators as it is equipped with standard built-in high-level libraries, including compatibility with SQL queries (SparkSQL), machine learning (MLlib), streaming data and graph processing (GraphX) in addition to the simple MapReduce functionalities of the MR model. It can be used in real-time processing applications by applying transformations to semi-structured data with the option of allowing interactive querying within the Spark shell. This dynamic nature has led to it being more popular than Hadoop.

Hadoop support: Big Data and the cloud are synergistic and Spark's support for cloud technologies is one of its biggest advantages. It is compatible with widely used Big Data frameworks like HDFS, Apache Cassandra, Apache Hbase, Apache Mesos and Amazon S3. Spark, which doesn't have its own storage system, enhances the Hadoop stack by implementing it in three possible ways: 1) standalone mode, 2) over YARN, or 3) SIMR (Spark in MapReduce). It can also support existing pure Hadoop ecosystems.

MapReduce alternative: Spark can be used instead of MapReduce as it executes jobs in short, micro bursts of 5 seconds or less. It is a faster framework for batch processing and iterative algorithms in comparison to Hadoop-based frameworks like Twitter Storm for live processing.

Configuring Apache Spark on Ubuntu

It is easy to install and configure Apache Spark on Ubuntu. A native Linux system is preferred as it provides the best environment for deployment. Virtual OSs can also be used, but the performance gets compromised when compared to the native versions. Dual OSs work satisfactorily. There are options to use a standalone version or use a version pre-built for Hadoop, which utilises the existing Hadoop components such as HDFS or a version built to be deployed on YARN. The following section will explain how to get Spark 2.0.0 standalone mode running on Ubuntu 14.04 or later.

Installing Java: To install and configure Spark, your machine needs Java. Use the following commands in a terminal to automatically download and update Java:

```
$ sudo apt-add-repository ppa:webupd8team/java
$ sudo apt-get update
$ sudo apt-get install oracle-java7-installer
```

You can check for an existing version by typing:

```
$ java -version
```

Installing Scala: Spark is written in Scala; so we need it to install the former. Download version 2.10.4 or later from <http://www.scala-lang.org/>.

Untar the file by using the following command:

```
$ sudo tar xvf scala-2.10.4.tgz
```

Add an entry for Scala in the file `.bashrc`, as follows:

```
nano ~/.bashrc
```

At the end of the file, add the path given below to show the location of the Scala file:

```
export SCALA_HOME=<path-where-scala-file-is-located>
export PATH=$SCALA_HOME/bin:$PATH
```

Then we need to source the changed `.bashrc` file by using the command given below:

```
source ~/.bashrc
```

We can verify the Spark installation by using the following command:

```
$ scala -version
```